

Lab 4: Disk Image and Partitions

What You Need for this lab

- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>
 - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- Disk image : https://www.dropbox.com/s/1fop1ooadb2yshu/Disk_Image_ID-20210327.001

1. Verify the integrity of the disk image

- Create Lab Folder

```
(ubalt@kali-forensics)-[~]  
$ mkdir Illegal_Download_Case  
mkdir: created directory 'Illegal_Download_Case'  
  
(ubalt@kali-forensics)-[~]  
$ cd Illegal_Download_Case  
  
(ubalt@kali-forensics)-[~/Illegal_Download_Case]  
$ pwd  
/home/ubalt/Illegal_Download_Case
```

Create lab folder in /home directory.

Change directory into the lab folder.

Verify that you are in the lab folder.

- Download Case Materials

```
(ubalt@kali-forensics)-[~/Illegal_Download_Case]  
$ mkdir Case_Materials  
mkdir: created directory 'Case_Materials'  
  
(ubalt@kali-forensics)-[~/Illegal_Download_Case]  
$ cd Case_Materials
```

Create folder that will contain disk image and memory file.

- Use *wget* to download disk image. (about 30GB)

```
(ubalt@kali-forensics)-[~/Illegal_Download_Case/Case_Materials]  
$ wget -q https://www.dropbox.com/s/1fop1ooadb2yshu/Disk_Image_ID-20210327.001
```

- Record Hash Information
- Open a text file using the text editor Nano:

```
(ubalt@kali-forensics)-[~/Illegal_Download_Case/Case_Materials]
$ nano hash_reference.txt
```

```
GNU nano 5.3 hash_reference.txt *
# Note: MD5 is listed first, then SHA1 second.

# MD5 and SHA1 for Disk Image
0adf1f182aab391d7042133bb55cf832 Disk_Image_ID-20210327.001
36186d7abc2ab6dfb783f0283e96566db8c0b31c Disk_Image_ID-20210327.001

# MD5 and SHA1 for Memory Dump
eb741d12f5d4f57d4c1d5962bebb5771 Memory_Dump_ID-20210327.raw
59bfe8d18c1e3dcd5a4ffe920adea39ec3257e87 Memory_Dump_ID-20210327.raw

# MD5 and SHA1 for Files Allegedly Download/Uploaded
f9d82438ed446e43cbd06adf74bc875b Contraband.mp3
cce250d81a2397a4b347b78337219f078dcd4e08 Contraband.mp3

b387e8334b3db0346cdfa14511f6913d Sample-1.mp3
23181f3d97a836106aae5db86b22bc34d4b1c138 Sample-1.mp3
```

This information will be referenced during the lab demonstration.

– Install Necessary Software

☐ Hashdeep

☐ Md5deep

sudo apt install hashdeep

Step 2.

Make sure you are in the lab folder

```
(ubalt@kali-forensics)-[~]
$ cd Illegal_Download_Case

(ubalt@kali-forensics)-[~/Illegal_Download_Case]
$
```

Generate an MD5 and SHA1 hash of the disk image. These tools will compare the MD5 and/or SHA1 hash of the disk image to the MD5 and/or SHA1 hash in the 'hash_reference.txt' file.

Commands

- `md5deep <disk image> -bewM <file that contains file names and hash codes>`
- `sha1deep <disk image> -bewM <file that contains file names and hash codes>`

- Note: You would replace <disk image> with the file path to the disk image. The same applies to anything else contained in between '<>'.
 - Use MD5deep to verify the MD5 hash of the disk image.

```
(ubalt@ kali-forensics)-[~/Illegal_Download_Case]
$ md5deep Case_Materials/Disk_Image_ID-20210327.001 -bewM Case_Materials/hash_reference.txt
md5deep: Case_Materials/hash_reference.txt: No hash found in line 1
md5deep: Case_Materials/hash_reference.txt: No hash found in line 2
md5deep: Case_Materials/hash_reference.txt: No hash found in line 3
md5deep: Case_Materials/hash_reference.txt: No hash found in line 5
md5deep: Case_Materials/hash_reference.txt: No hash found in line 6
md5deep: Case_Materials/hash_reference.txt: No hash found in line 7
md5deep: Case_Materials/hash_reference.txt: No hash found in line 9
md5deep: Case_Materials/hash_reference.txt: No hash found in line 10
md5deep: Case_Materials/hash_reference.txt: No hash found in line 11
md5deep: Case_Materials/hash_reference.txt: No hash found in line 13
md5deep: Case_Materials/hash_reference.txt: No hash found in line 14
md5deep: Case_Materials/hash_reference.txt: No hash found in line 16
md5deep: Case_Materials/hash_reference.txt: No hash found in line 17
md5deep: Case_Materials/hash_reference.txt: No hash found in line 18
0adf1f182aab391d7042133bb55cf832 Disk_Image_ID-20210327.001 matched Disk_Image_ID-20210327.001
```

The MD5 of the disk image matches the one given in hash_reference.txt.

- Use SHA1deep to verify the MD5 hash of the disk image.

```
(ubalt@ kali-forensics)-[~/Illegal_Download_Case]
$ sha1deep Case_Materials/Disk_Image_ID-20210327.001 -bewM Case_Materials/hash_reference.txt
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 1
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 2
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 3
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 4
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 6
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 7
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 8
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 10
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 11
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 12
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 14
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 15
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 17
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 18
36186d7abc2ab6dfb783f0283e96566db8c0b31c Disk_Image_ID-20210327.001 matched Disk_Image_ID-20210327.001
```

The SHA1 of the disk image matches the one given in hash_reference.txt.

2. Identify the OS of the system as well as its name, accounts, and partitions.

- How to get help for *fdisk*.

```
(ubalt@ kali-forensics)-[~/Illegal_Download_Case]
$ fdisk -h

Usage:
fdisk [options] <disk>          change partition table
fdisk [options] -l [<disk> ...] list partition table(s)

Display or manipulate a disk partition table.

Options:
-b, --sector-size <size>      physical and logical sector size
-B, --protect-boot            don't erase bootbits when creating a new label
-c, --compatibility[=<mode>]  mode is 'dos' or 'nondos' (default)
-L, --color[=<when>]          colorize output (auto, always or never)
                              colors are enabled by default
-l, --list                    display partitions and exit
-x, --list-details            like --list but with more details
-n, --noauto-pt              don't create default partition table on empty devices
-o, --output <list>          output columns
-t, --type <type>            recognize specified partition table type only
-u, --units[=<unit>]          display units: 'cylinders' or 'sectors' (default)
-s, --getsz                  display device size in 512-byte sectors [DEPRECATED]
                              print SIZE in bytes rather than in human readable format
                              --lock[=<mode>] use exclusive device lock (yes, no or nonblock)
                              --wipe <mode> wipe signatures (auto, always or never)
                              -W, --wipe-partitions <mode> wipe signatures from new partitions (auto, always or never)

-C, --cylinders <number>     specify the number of cylinders
-H, --heads <number>         specify the number of heads
-S, --sectors <number>       specify the number of sectors per track

-h, --help                    display this help
-V, --version                 display version
```

- Use *fdisk* to get the disk image's partition table.

```
(ubalt@ kali-forensics) - [~/Illegal_Download_Case]
$ cd Case_Materials
```

Change into the Case_Materials folder

```
(ubalt@ kali-forensics) - [~/Illegal_Download_Case/Case_Materials]
```

Note: If you do not know where you are, this will tell you.

```
(ubalt@ kali-forensics) - [~/Illegal_Download_Case/Case_Materials]
$ fdisk -l Disk Image ID-20210327.001
```

Disk Image ID-20210327.001: 30 GiB, 32212254720 bytes, 62914560 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8afa8be3

Device	Boot	Start	End	Sectors	Size	Id	Type
Disk Image ID-20210327.001p1	*	2048	104447	102400	50M	7	HPFS/NTFS/exFAT
Disk Image ID-20210327.001p2		104448	61890501	61786054	29.5G	7	HPFS/NTFS/exFAT
Disk Image ID-20210327.001p3		61890560	62910463	1019904	498M	27	Hidden NTFS WinRE

Volume offset #s (in sectors):

- Partition 1 – 2048
 - Partition 2 – 104448
 - Partition 3 – 61890560
- How to get help for *fsstat*

```
(ubalt@ kali-forensics) - [~/Illegal_Download_Case/Case_Materials]
$ fsstat
```

Missing image name
usage: fsstat [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image
-t: display type only
-i imgtype: The format of the image file (use '-i list' for supported types)
-b dev_sector_size: The size (in bytes) of the device sectors
-f fstype: File system type (use '-f list' for supported types)
-o imgoffset: The offset of the file system in the image (in sectors)
-P pooltype: Pool container type (use '-P list' for supported types)
-B pool_volume_block: Starting block (for pool volumes only)
-v: verbose output to stderr
-V: Print version
-k password: Decryption password for encrypted volumes

```
(ubalt@ kali-forensics) - [~/Illegal_Download_Case/Case_Materials]
$
```

- Use *fsstat* to get file system details.

```
(ubalt@ kali-forensics)-[~/Illegal_Download_Case/Case_Materials]
$ fsstat -o 2048 Disk Image ID-20210327.001
FILE SYSTEM INFORMATION
```

```
File System Type: NTFS
Volume Serial Number: 18EC42BBEC4292C4
OEM Name: NTFS
Volume Name: System Reserved
Version: Windows XP
```

```
(ubalt@ kali-forensics)-[~/Illegal_Download_Case/Case_Materials]
$ fsstat -o 104448 Disk Image ID-20210327.001
FILE SYSTEM INFORMATION
```

```
File System Type: NTFS
Volume Serial Number: E8DE4350DE4315EA
OEM Name: NTFS
Version: Windows XP
```

```
(ubalt@ kali-forensics)-[~/Illegal_Download_Case/Case_Materials]
$ fsstat -o 61890560 Disk Image ID-20210327.001
FILE SYSTEM INFORMATION
```

```
File System Type: NTFS
Volume Serial Number: 9E46F86046F83A9B
OEM Name: NTFS
Version: Windows XP
```

Partition 1

File System: ?

Serial Number: ?

Partition 2

File System: ?

Serial Number: ?

Partition 3

File System: ?

Serial Number: ?

- Using *fdisk* and *fsstat*, we obtained this information:

Partition Table			MS-DOS				
Partition	Flag	Start	End	Sectors	Size	File System	Serial #
1st Partition – System Reserved	Boot	2048	104447	102400	50 MB	NTFS	18EC42BBEC4292C4
2nd Partition	-	104448	61890501	61786054	29.5 GB	NTFS	E8DE4350DE4315EA
3rd Partition	-	61890560	62910463	1019904	498 MB	NTFS/Hidden NTFS WinRe	9E46F86046F83A9B

Please explain the parameters in the table ?

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 4.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 4 From YOUR NAME**", replacing "YOUR NAME" with your real name.